Prosit 6

1a-i

```
SQL> set serveroutput on ;
SQL> DECLARE
    v_dept_id     departments.department_id%TYPE;
    v_dept_name   departments.department_name%TYPE;
    v_avg_sal     NUMBER;
  BEGIN
    -- Curseur implicite : FOR ... IN (SELECT ...)
    FOR rec IN (
      SELECT d.department_id,
             d.department_name,
             ROUND(AVG(e.salary), 2) AS avg_salary
      FROM   departments d
      JOIN   employees   e ON d.department_id = e.department_id
      GROUP BY d.department_id, d.department_name
      ORDER BY d.department_id
    ) LOOP
      DBMS_OUTPUT.PUT_LINE(
        'ID: '     || rec.department_id   ||
        '  Nom: ' || rec.department_name ||
        '  Sal. moy: ' || rec.avg_salary
      );
    END LOOP;
  END;
  /
ID: 10  Nom: Administration  Sal. moy: 4400
ID: 20  Nom: Marketing  Sal. moy: 9500
ID: 30  Nom: Purchasing  Sal. moy: 4150
ID: 40  Nom: Human Resources  Sal. moy: 6500
ID: 50  Nom: Shipping  Sal. moy: 3475,56
ID: 60  Nom: IT  Sal. moy: 5760
ID: 70  Nom: Public Relations  Sal. moy: 10000
ID: 80  Nom: Sales  Sal. moy: 8955,88
ID: 90  Nom: Executive  Sal. moy: 19333,33
ID: 100  Nom: Finance  Sal. moy: 8601,33
ID: 110  Nom: Accounting  Sal. moy: 10154

PL/SQL procedure successfully completed.
```

1a-ii

```
SQL> DECLARE
    -- Déclaration du curseur explicite
    CURSOR c_dept IS
      SELECT d.department_id,
             d.department_name,
             ROUND(AVG(e.salary), 2) AS avg_salary
      FROM   departments d
      JOIN   employees   e ON d.department_id = e.department_id
      GROUP BY d.department_id, d.department_name
      ORDER BY d.department_id;

    v_dept_id     departments.department_id%TYPE;
    v_dept_name   departments.department_name%TYPE;
    v_avg_sal     NUMBER;
  BEGIN
    OPEN c_dept;
    LOOP
      FETCH c_dept INTO v_dept_id, v_dept_name, v_avg_sal;
      EXIT WHEN c_dept%NOTFOUND;
      DBMS_OUTPUT.PUT_LINE(
        'ID: '     || v_dept_id   ||
        '  Nom: ' || v_dept_name ||
        '  Sal. moy: ' || v_avg_sal
      );
    END LOOP;
    CLOSE c_dept;
  END;
  /
ID: 10  Nom: Administration  Sal. moy: 4400
ID: 20  Nom: Marketing  Sal. moy: 9500
ID: 30  Nom: Purchasing  Sal. moy: 4150
ID: 40  Nom: Human Resources  Sal. moy: 6500
ID: 50  Nom: Shipping  Sal. moy: 3475,56
ID: 60  Nom: IT  Sal. moy: 5760
ID: 70  Nom: Public Relations  Sal. moy: 10000
ID: 80  Nom: Sales  Sal. moy: 8955,88
ID: 90  Nom: Executive  Sal. moy: 19333,33
ID: 100  Nom: Finance  Sal. moy: 8601,33
ID: 110  Nom: Accounting  Sal. moy: 10154

PL/SQL procedure successfully completed.
```

1b

```
SQL> BEGIN
    FOR rec IN (
      SELECT d.department_id,
             d.department_name,
             COUNT(e.employee_id) AS nb_emp
      FROM   departments d
      LEFT JOIN employees e ON d.department_id = e.department_id
      GROUP BY d.department_id, d.department_name
      ORDER BY d.department_id
    ) LOOP
      DBMS_OUTPUT.PUT_LINE(
        'Département ' || rec.department_name ||
        ' (ID ' || rec.department_id || ') : ' ||
        rec.nb_emp || ' employé(s)'
      );
    END LOOP;
  END;
  /
Département Administration (ID 10) : 1 employé(s)
Département Marketing (ID 20) : 2 employé(s)
Département Purchasing (ID 30) : 6 employé(s)
Département Human Resources (ID 40) : 1 employé(s)
Département Shipping (ID 50) : 45 employé(s)
Département IT (ID 60) : 5 employé(s)
Département Public Relations (ID 70) : 1 employé(s)
Département Sales (ID 80) : 34 employé(s)
Département Executive (ID 90) : 3 employé(s)
Département Finance (ID 100) : 6 employé(s)
Département Accounting (ID 110) : 2 employé(s)
Département Treasury (ID 120) : 0 employé(s)
Département Corporate Tax (ID 130) : 0 employé(s)
Département Control And Credit (ID 140) : 0 employé(s)
Département Shareholder Services (ID 150) : 0 employé(s)
Département Benefits (ID 160) : 0 employé(s)
Département Manufacturing (ID 170) : 0 employé(s)
Département Construction (ID 180) : 0 employé(s)
Département Contracting (ID 190) : 0 employé(s)
Département Operations (ID 200) : 0 employé(s)
Département IT Support (ID 210) : 0 employé(s)
Département NOC (ID 220) : 0 employé(s)
Département IT Helpdesk (ID 230) : 0 employé(s)
Département Government Sales (ID 240) : 0 employé(s)
Département Retail Sales (ID 250) : 0 employé(s)
Département Recruiting (ID 260) : 0 employé(s)
Département Payroll (ID 270) : 0 employé(s)

PL/SQL procedure successfully completed.
```

1c

```
SQL> DECLARE
    CURSOR c_emp_big_dept IS
      SELECT e.employee_id,
             e.first_name,
             e.last_name,
             d.department_name,
             COUNT(e2.employee_id) OVER
               (PARTITION BY d.department_id) AS nb_emp
      FROM   employees   e
      JOIN   departments d  ON e.department_id = d.department_id
      JOIN   employees   e2 ON e2.department_id = e.department_id
      GROUP BY e.employee_id, e.first_name, e.last_name,
               d.department_id, d.department_name
      HAVING COUNT(e2.employee_id) > 20
      ORDER BY d.department_name, e.last_name;

    -- Variante plus simple avec sous-requête
    CURSOR c_simple IS
      SELECT e.employee_id, e.first_name, e.last_name,
             d.department_name
      FROM   employees e
      JOIN   departments d ON e.department_id = d.department_id
      WHERE  e.department_id IN (
               SELECT department_id
               FROM   employees
               GROUP BY department_id
               HAVING COUNT(*) > 20
             )
      ORDER BY d.department_name, e.last_name;

    v_rec c_simple%ROWTYPE;
  BEGIN
    OPEN c_simple;
    LOOP
      FETCH c_simple INTO v_rec;
      EXIT WHEN c_simple%NOTFOUND;
      DBMS_OUTPUT.PUT_LINE(
        v_rec.last_name || ' ' || v_rec.first_name ||
        ' - Dept: ' || v_rec.department_name
      );
    END LOOP;
    CLOSE c_simple;
  END;
  /
```

```
Abel Ellen - Dept: Sales
Ande Sundar - Dept: Sales
Banda Amit - Dept: Sales
Bates Elizabeth - Dept: Sales
Bernstein David - Dept: Sales
Bloom Harrison - Dept: Sales
Cambrault Gerald - Dept: Sales
Cambrault Nanette - Dept: Sales
Doran Louise - Dept: Sales
Errazuriz Alberto - Dept: Sales
Fox Tayler - Dept: Sales
Greene Danielle - Dept: Sales
Hall Peter - Dept: Sales
Hutton Alyssa - Dept: Sales
Johnson Charles - Dept: Sales
King Janette - Dept: Sales
Kumar Sundita - Dept: Sales
Lee David - Dept: Sales
Livingston Jack - Dept: Sales
Marvins Mattea - Dept: Sales
McEwen Allan - Dept: Sales
Olsen Christopher - Dept: Sales
Ozer Lisa - Dept: Sales
Partners Karen - Dept: Sales
Russell John - Dept: Sales
Sewall Sarath - Dept: Sales
Smith Lindsey - Dept: Sales
Smith William - Dept: Sales
Sully Patrick - Dept: Sales
Taylor Jonathon - Dept: Sales
Tucker Peter - Dept: Sales
Tuvault Oliver - Dept: Sales
Vishney Clara - Dept: Sales
Zlotkey Eleni - Dept: Sales
Atkinson Mozhe - Dept: Shipping
Bell Sarah - Dept: Shipping
Bissot Laura - Dept: Shipping
Bull Alexis - Dept: Shipping
```

1d

```
SQL> DECLARE
    CURSOR c_managers IS
      SELECT DISTINCT
             m.employee_id,
             m.first_name AS mgr_first,
             m.last_name  AS mgr_last
      FROM   employees m
      WHERE  m.employee_id = 100;   -- uniquement le manager 100

    CURSOR c_employes (p_mgr_id employees.manager_id%TYPE) IS
      SELECT first_name, last_name
      FROM   employees
      WHERE  manager_id = p_mgr_id
      ORDER BY last_name;

    v_emp c_employes%ROWTYPE;
  BEGIN
    FOR mgr IN c_managers LOOP
      DBMS_OUTPUT.PUT_LINE(
        '=== Manager : ' || mgr.mgr_last || ' ' || mgr.mgr_first || ' ==='
      );
      OPEN c_employes(mgr.employee_id);
      LOOP
        FETCH c_employes INTO v_emp;
        EXIT WHEN c_employes%NOTFOUND;
        DBMS_OUTPUT.PUT_LINE(
          '    > ' || v_emp.last_name || ' ' || v_emp.first_name
        );
      END LOOP;
      CLOSE c_employes;
    END LOOP;
  END;
  /
=== Manager : King Steven ===
> Cambrault Gerald
> De Haan Lex
> Errazuriz Alberto
> Fripp Adam
> Hartstein Michael
> Kaufling Payam
> Kochhar Neena
> Mourgos Kevin
> Partners Karen
> Raphaely Den
> Russell John
> Vollman Shanta
> Weiss Matthew
> Zlotkey Eleni
```

2

```
SQL> BEGIN
    -- DDL dynamique obligatoire dans un bloc PL/SQL
    EXECUTE IMMEDIATE '
      CREATE TABLE PRODUITS (
        idProduit     NUMBER(10)      CONSTRAINT pk_produits PRIMARY KEY,
        nomProduit    VARCHAR2(100)   NOT NULL,
        categorieP    VARCHAR2(50),
        prixProduit   NUMBER(10, 2)   NOT NULL
      )
    ';
    DBMS_OUTPUT.PUT_LINE('Table PRODUITS créée avec succès.');

  EXCEPTION
    WHEN OTHERS THEN
      -- ORA-00955 : table already exists
      IF SQLCODE = -955 THEN
        DBMS_OUTPUT.PUT_LINE('La table PRODUITS existe déjà.');
      ELSE
        DBMS_OUTPUT.PUT_LINE('Erreur : ' || SQLERRM);
      END IF;
  END;
  /
Table PRODUITS créée avec succès.

PL/SQL procedure successfully completed.
```